Memory Based Questions of BITS HD (26th May, 2026)

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Gate score:- 442
Bits Hd:- 85

There were two slots first paper was from 25th May. I was in second slot. The test paper consist of two paper paper 1 and 2. Test 1 was of Logical reasoning and English (15 questions) and maths (15 questions). Test 2 was of 70 questions both electrical and electronics combined paper comes. Correct answer +3 and incorrect answer -1. Calculator is allowed.

TEST 1: -

PROPERTY BASED QUESTION OF LINEAR ALGEBRA AND COMPLEX VARIABLE:-

1. If A matrix is invertible then :- a. eigne value 0 b. eigen value not zero c. independent matrix d. dependent matrix
2. If eigen value of A is 2, 4 and 6 and eigen value of B is 0, 1, 3 then which of the following equation satisfy:-
   a. A – 2I b. 3A – 3I c. A- 3i
3. Which is incorrect:-
   a. |z1 + z2| >= |z1| + |z2|
   b. |z1 - z2| >= |z1| - |z2|
   c. |z1 + z2| <= |z1| + |z2|
   d. |z1 - z2| <= |z1| - |z2|

Basics of differential equations: -

4. Order and degree question
5. Probability question of normal distribution
6. Calculus question

$$\int_{1}^{-1} \frac{x^2}{\sqrt{1-x^2}}\, dx$$

$$e^{-2s}\sin^2\theta :$$

7. Laplace question of
8. Find Polar arc of r = 2sin theta(don't know the exact question)

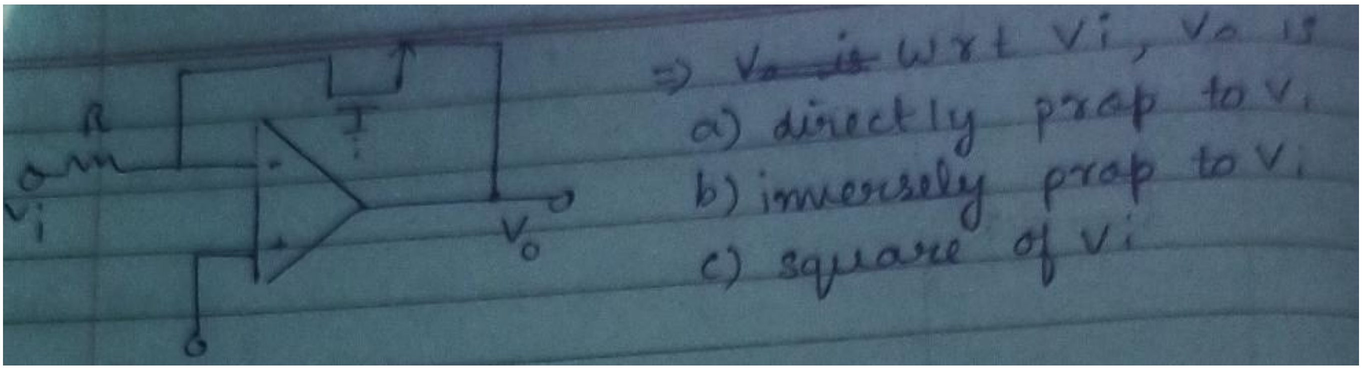
$$\frac{1}{z^2-1}$$

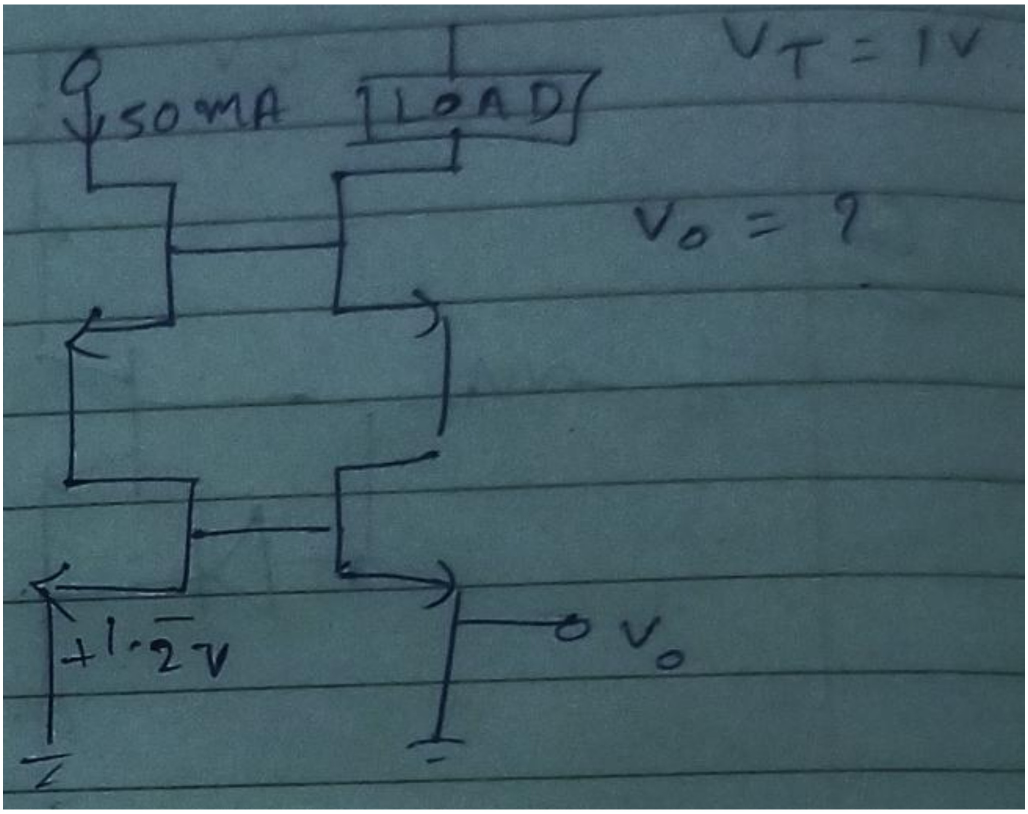
9. Complex calculus contour was \z\=2 then in positive orientation what will be
   a. 2pi I b. 1 c. 0 d. -1
10. Which of the following is analytical everywhere:-
    a. e^sinz b|z| c. $\bar{z}\, z$:

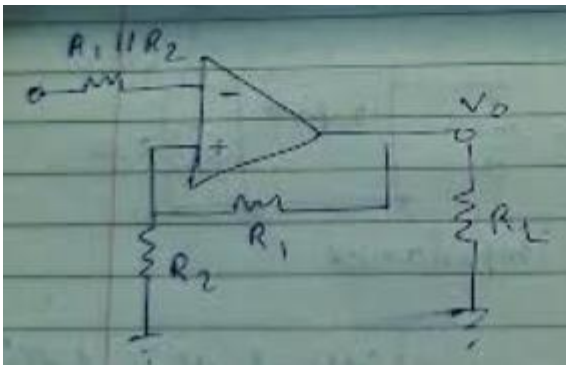
(Other miscellaneous questions were also there all from basics. Not over the top but basic)

TEST 2: -

1. Most of the questions were from Analog Electronics you should have good knowledge about the diode bjt and opamp. Mostly the dc suppy of opamp was the favourite part of bits. If one wants to score marks in bits he or she should be very good in analog part logic wise)
1. Electrical machines power electronics had questions like 20 25.
2. In edc there were pretty much basic questions (some should have good knowledge about the pn junction working and their carriers information what happens something when something is increased or decreased)
3. EDC:-
   1. What will the fermi level separation if electron concentration is 10^16 and holes concentration Is 10^17. Ni = 1.5 * 10^10. A. 0.407 and 0.397 b. 0.397 and 0.407 etc.
   2. Where the depletion region width will be more if the doping at n side is 2* 10^14 and doping at p side is 9* 10 ^17? A. p side will have more and side smal b. n side small p side big d. both will have equal length of depletion region.
   3. If we have p+n and the current is responsible due to injection of holes in n side and electrons in p side then what you will do to increase the current of the silicon: a. injection of holes in n side and increase doping in p side . b. injection of electrons in p side and reduce doping in n side etc.
   4. If external potential is not given then what will be the built in potential? A. vtln (NaNd/ni^2)

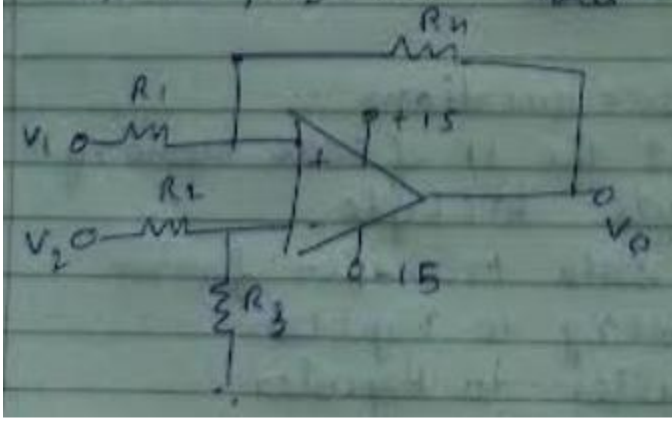




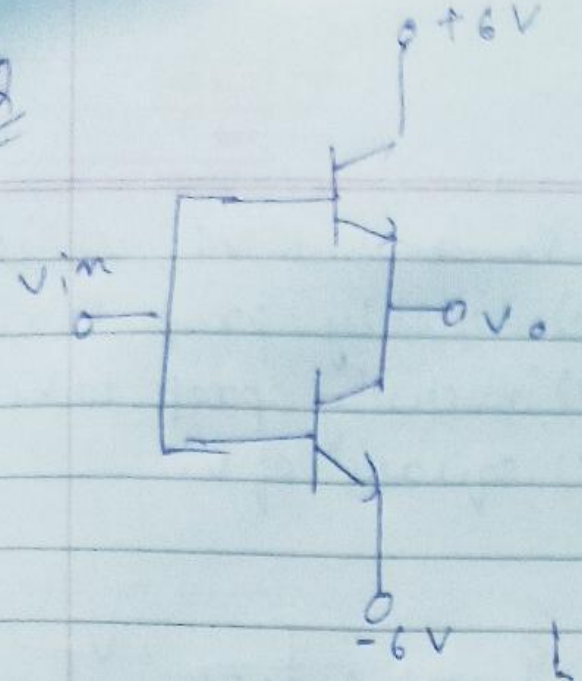


Identify which ckt
a) Schmitt
b) square wave
c) pulsating
d) phasor

$V_1 = 1V$, $V_2 = 2V$ $V_{out} = ?$ R_1, R_2 given

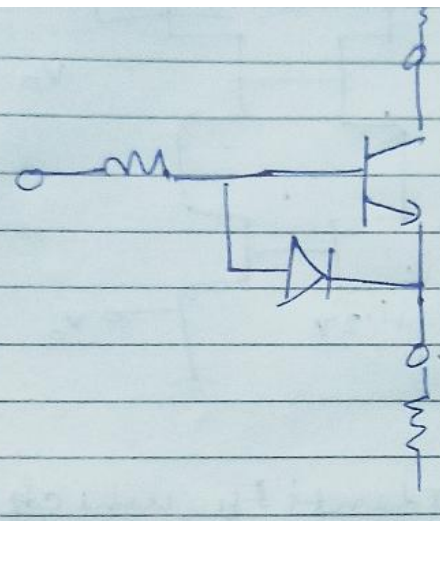


Q



$V_{in} = 4.5V$ p-p.

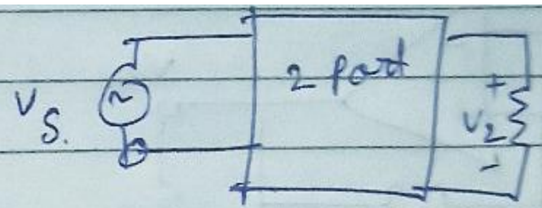
power = ?



Something like this

Q $Z_{11} = Z_{22} = 10\Omega$

$Z_{21} = Z_{12} = 4\Omega$.



What is input impedance.

Q. $\frac{K}{(S+1)^2}, \frac{K}{(S+1)^4}, \frac{K}{(S+1)^3}$, which of the folling is more stable?

Micro processor questions:-

1) What is the IP of 8086 memory?
   a) 4 byte b) 6 byte
2) 8085 Data transfer from
   a) Memory to Register
   b) Register to Register
   c) Register to Memory
   d) All of these.

3) 8086 which of the folling is in BIU
   a) ES b) SS c) BP d) IP

4) 8086 the CS segment register is of how much
   a) 1 MB b) 64 kb c) 84 bits

5) 8086 segment has 64KB then from 20000H, ending?.
   a) 2FFFFH b) 02FFFH c) 2FFF0H.

1) 10 MHz clock → Mod 5, Mod 8, Mod 10, final clock pulse.
   a) 25 KHz b) 10 MHz c) 2.5 KHz

2) $b_0, b_1, b_2, b_3 \rightarrow$ BCD code $\rightarrow$ if o/p is 1 when number is odd and less than 6 and greater the 7. Write essential prime implicants.

a) $\overline{B_1}B_0$ and $B_2\overline{B_3}$ b) $\overline{B_0}B_1$ or $\overline{B_2}B_3$